1. Regression updated

Regression analysis allows us to estimate the relationship of a response variable to a set of predictor variables

$y = mx + c$

Let

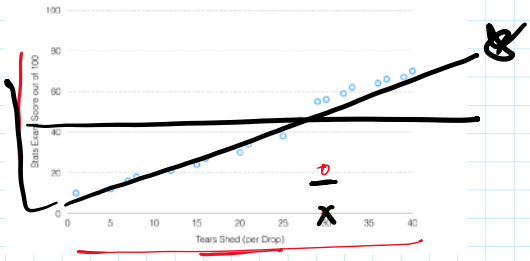
$x_1, x_2, \cdots x_n$ be settings of x chosen by the investigator and

$y_1, y_2, \cdots y_n$ be the corresponding values of the response.

Assume y_i is an observation of rv Y_i (which depends on x_i, where x_i is not random).

We model each Y_i by

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$



Regression Analysis

- Regression analysis is a statistical method to model the relationship between a dependent (target) and independent (predictor) variables with one or more independent variables.
- More specifically, Regression analysis helps us to understand how the value of the dependent variable is changing corresponding to an independent variable when other independent variables are held fixed.
- It predicts continuous/real values such as temperature, age, salary, price, etc.

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

Regression Analysis

Suppose there is a marketing company A, who does various advertisement every year and get sales on that. The below list shows the advertisement made by the company in the last 5 years and the corresponding sales:

Advertisement	Sales
$90	$1000
$120	$1300
$150	$1800
$100	$1200
$130	$1380
$200	??

Regression Analysis

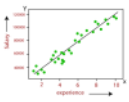
- Regression is a supervised learning technique which helps in finding the correlation between variables and enables us to predict the continuous output variable based on the one or more predictor variables.
- It is mainly used for prediction, forecasting, time series modeling, and determining the causal-effect relationship between variables.
- *"Regression shows a line or curve that passes through all the datapoints on target-predictor graph in such a way that the vertical distance between the datapoints and the regression line is minimum."*
- The distance between datapoints and line tells whether a model has captured a strong relationship or not.

Linear Regression

- Linear regression is a statistical regression method which is used for predictive analysis.

Linear regression shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis), hence called linear regression.

If there is only one input variable (x), then such linear regression is called **simple linear regression**. And if there is more than one input variable, then such linear regression is called **multiple linear regression**



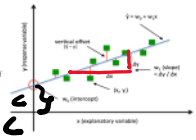
Simplest Possible Linear Regression Model

- This is the base model for all statistical machine learning
- x is a one feature data variable
- y is the value we are trying to predict
- The regression model is

$$y = w_0 + w_1 x + \epsilon$$

Two parameters to estimate – the slope of the line w_1 and the y-intercept w_0

- ϵ is the unexplained, random, or error component.



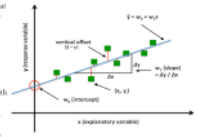
$y = mx + c$

Solving the regression problem

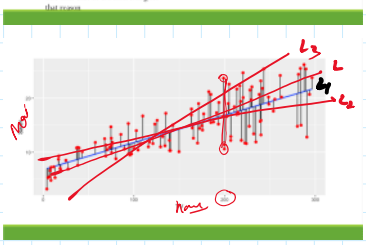
- We basically want to find $\{w0, w1\}$ that minimize deviations from the predictor line:

$$\arg\min_{w_0, w_1} \sum_i^n (y_i - w_0 - w_1 x_i)^2$$

- How do we do it?
  - Iterate over all possible w values along the two dimensions?
  - No, we can do this in closed form with just plain calculus
- Very few optimization problems in ML have closed form solutions
  - The ones that do are interesting for that reason



- The ones that do are interesting for that reason



Regression. (Sir Francis Galton) (1)

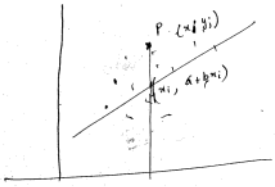
(Stepping back towards the average)

- If two R.V's X & Y are related, then we will try to find some curve in scatter diagram defined as "Curve of regression", which provides mathematical measure of average relationship between two or more R.V.s.

- The line of Regression gives best estimate for the value of one variable for any specific value of other variable.

- Regression ~~principle~~ is based on "principle of least squares"

$\hat{y}_i = a + bx_i$



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- Legendre's principle of Least square, consists of minimising the sum of the squares of the (error) estimate deviations of the actual values of Y from their estimated values as given by line of best fit (L.O.R.)

$E = (y_i - \hat{y}_i)$

$\hat{y}_i$ = estimated value

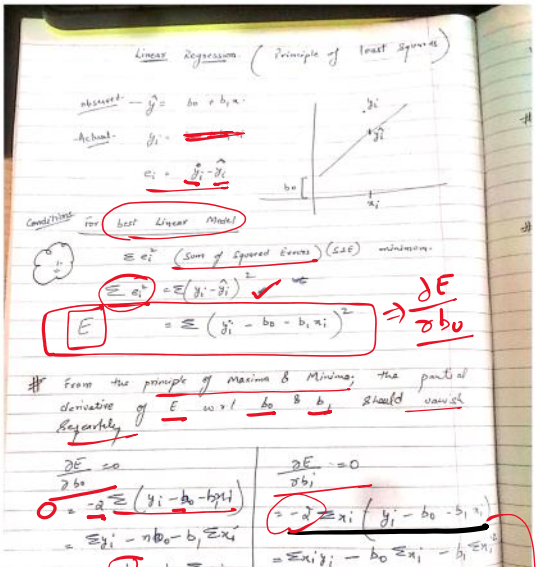
$\boxed{E} = (y_i - \hat{y}_i)$

$E = y_i - (a + bx_i)$

(2) $\hat{y}_i = a + bx_i$

$\boxed{\hat{y}_i}$ $\hat{y}_j$ y_i x_i

$\Sigma(y_j - \hat{y}_j)$



Linear Regression (Principle of least Squares)

observed — $\hat{y}_i = b_0 + b_1 x_i$

Actual — y_i

$e_i = y_i - \hat{y}_i$

Conditions for best Linear Model

Σe_i^2 (Sum of Squared Errors) (SSE) minimum.

$\Sigma e_i^2 = \Sigma(y_i - \hat{y}_i)^2$

$\boxed{E} = \Sigma(y_i - b_0 - b_1 x_i)^2 \Rightarrow \frac{\partial E}{\partial b_0}$

From the principle of Maxima & Minima, the partial derivative of E w.r.t b_0 & b_1 should vanish.

Separately

$\frac{\partial E}{\partial b_0} = 0$

$0 = -2\Sigma(y_i - b_0 - b_1 x_i)$

$= \Sigma y_i - nb_0 - b_1 \Sigma x_i$

$\frac{\partial E}{\partial b_1} = 0$

$= -2\Sigma x_i(y_i - b_0 - b_1 x_i)$

$= \Sigma x_i y_i - b_0 \Sigma x_i - b_1 \Sigma x_i^2$

$\Sigma y_i - \Sigma b_0 - \Sigma b_1 x_i = 0$

$\Sigma y_i - nb_0 - b_1 \Sigma x_i = 0$ (1)

$$\frac{\partial}{\partial b_0} = -2\sum(y_i - b_0 - b_1 x_i)$$

$$= \sum y_i - nb_0 - b_1\sum x_i$$

$$\sum y_i = nb_0 + b_1\sum x_i \quad (1)$$

$$\frac{\partial}{\partial b_1} = -2\sum x_i(y_i - b_0 - b_1 x_i)$$

$$= \sum x_i y_i - b_0\sum x_i - b_1\sum x_i^2$$

$$\sum x_i y_i = b_0\sum x_i + b_1\sum x_i^2 \quad (2)$$

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$$\sum y_i - nb_0 - b_1\sum x_i = 0 \quad (1)$$

$$\bar{y} - b_0 - b_1\bar{x} = 0$$

$$\boxed{\bar{y} - b_1\bar{x} = b_0}$$

$$\sum x_i(y_i - b_0 - b_1 x_i) = 0$$

$$\sum x_i y_i - b_0\sum x_i - b_1\sum x_i^2 = 0$$

$$\sum x_i y_i - \frac{\left(\sum y_i - b_1\sum x_i\right)}{n}\sum x_i - b_1\sum x_i^2 = 0$$

$$\boxed{n\sum x_i y_i - \sum x_i y_i + b_1\sum x_i^2 - nb_1\sum x_i^2 = 0}$$

Value of b_0 & b_1 from (1) & (2) we can get

\# divide (1) / n

$$\boxed{\bar{y} = b_0 + b_1\bar{x}}$$

\# divide (2) / n

$$\frac{1}{n}\sum x_i y_i = b_0\bar{x} + \frac{b_1\sum x_i^2}{n}$$

put value of b_0

$$b_0 = \bar{y} - b_1\bar{x}$$

$$\frac{1}{n}\sum x_i y_i = (\bar{y} - b_1\bar{x})\bar{x} + b_1\frac{1}{n}\sum x_i^2$$

$$\frac{1}{n}\sum x_i y_i = \bar{x}\bar{y} - b_1\bar{x}^2 + b_1\frac{1}{n}\sum x_i^2$$

$$\frac{1}{n}\sum x_i y_i - \bar{x}\bar{y} = b_1\left(\frac{1}{n}\sum x_i^2 - \bar{x}^2\right)$$

$$b_1 = \boxed{\frac{\frac{1}{n}\sum x_i y_i - \bar{x}\bar{y}}{\frac{1}{n}\sum x_i^2 - \bar{x}^2}} \Leftrightarrow \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} \quad \frac{Cov(x,y)}{\sigma_x^2}$$

from this we can get value of b_0

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$$b_1 = \frac{Cov(x,y)}{\sigma_x^2}$$

$$Cov(x,y) = \sum(x_i - \bar{x})(y_i - \bar{y})$$

$$= \sum x_i y_i - \frac{1}{n}\sum x_i y_i$$

$$\sigma_x^2 = \sum(x_i - \bar{x})^2$$

$$= \sum x_i^2 - \frac{1}{n}\sum x_i^2$$

$$b_1 = \frac{Cov(x,y)}{\sigma_x^2}$$

(4) $\hat{y}_i = b_0 + b_1 x_i$

$y_i = a + bx_i$

Line of Regression of Y to X

$$y = \bar{y} + \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}(x - \bar{x})$$

Coeff of Regression for Y to X

$x_i = a + by_i$

the similarly we can get line of regression of X on Y

$$\hat{x} = \bar{x} + \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^{n} y_i^2 - n\bar{y}^2}(y - \bar{y})$$

Coeff of Regression for X to Y.

$$(y - \bar{y}) = \eta(x - \bar{x})$$

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Example for Linear Regression

$$b = \frac{n\sum xy - \sum x * \sum y}{n\sum x^2 - (\sum x)^2}$$

$$a = \frac{\sum y * \sum x^2 - \sum x * \sum xy}{n\sum x^2 - (\sum x)^2}$$

Find linear regression equation for the following two sets of data:

x	2	4	6	8
y	3	7	5	10

		x^2	xy
2	3	4	6
4	7	16	28
6	5	36	30
8	10	64	80
$\sum x = 20$	$\sum y = 25$	$\sum x^2 = 120$	$\sum xy = 144$

$b = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$

$b = \frac{4 \times 144 - 20 \times 25}{4 \times 120 - 400}$

b = 0.95

$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n\sum x^2 - (\sum x)^2}$

$a = \frac{25 \times 120 - 20 \times 144}{4 \times 120 - 400}$

a = 1.5

Linear regression is given by:

y = a + bx

y = 1.5 + 0.95 x

Evaluation Metrics for Regression Model

Mean Squared Error(MSE): MSE is a most used and very simple metric with a little bit of change in mean absolute error. Mean squared error states that finding the squared difference between actual and predicted value.

$$MSE = \frac{1}{n}\Sigma\left(y - \hat{y}\right)^2$$

$$E = \frac{1}{n}(y_i - \hat{y}_i)^2$$

Advantages of MSE
- The graph of MSE is differentiable, so you can easily use it as a loss function.

Disadvantages of MSE
- The value you get after calculating MSE is a squared unit of output. for example, the output variable is in meter(m) then after calculating MSE the output we get is in meter squared.
- If you have outliers in the dataset then it penalizes the outliers most and the calculated MSE is bigger. So, in short, it is not Robust to outliers which were an advantage in MAE.

Evaluation Metrics for Regression Model

Mean Absolute Error(MAE): calculates the absolute difference between actual and predicted values, so, sum all the errors and divide them by a total number of observations is MAE. And we aim to get a minimum MAE because this is a loss.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

where,

n *number of observation*

y_i *the actual value of the* i^{th} *observation*

$\hat{y}_i$ *the predicted value of the* i^{th} *observation*

Advantages of MAE

The MAE you get is in the same unit as the output variable.
It is more Robust to outliers.

Disadvantages of MAE

The graph of MAE is not differentiable so we have to apply various other optimizers.

Evaluation Metrics for Regression Model

R Squared (R2) R2 score is a metric that tells the performance of your model, not the loss in an absolute sense that how many wells did your model perform.

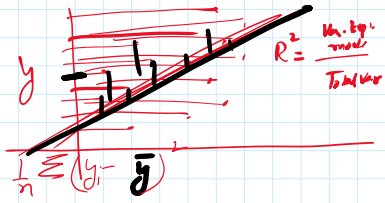
In contrast, MAE and MSE depend on the context as we have seen whereas the R2 score is independent of context.

R2 Squared = $1 - \frac{SSr}{SSm}$

SSr = Squared sum error of regression line

SSm = Squared sum error of mean line

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_i)^2}$$



Evaluation Metrics for Regression Model

Root Mean Squared Error(RMSE): As RMSE is clear by the name itself, that it is a simple square root of mean squared error.

$$RMSE = \sqrt{MSE}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Advantages of MSE
- The output value you get is in the same unit as the required output variable which makes interpretation of loss easy.

Disadvantages of RMSE
- It is not that robust to outliers as compared to MAE.
- Most of the time people use RMSE as an evaluation metric and mostly when you are working with deep learning techniques the most preferred metric is RMSE.

2. GD Regression

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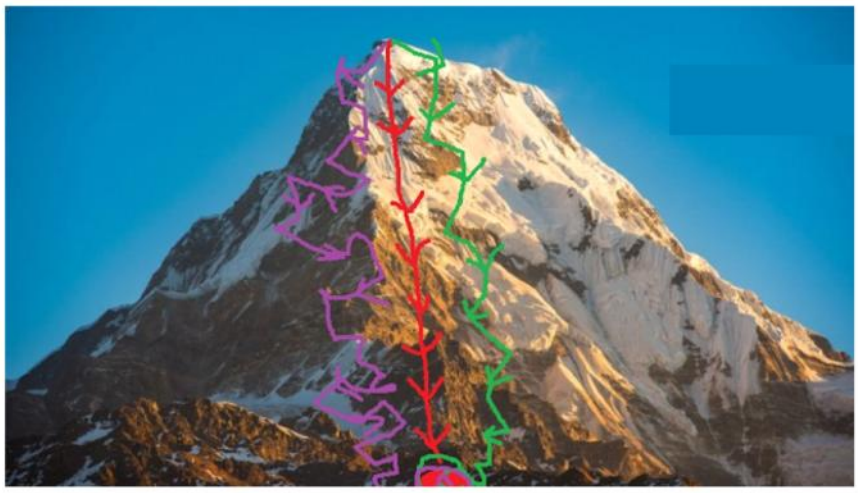
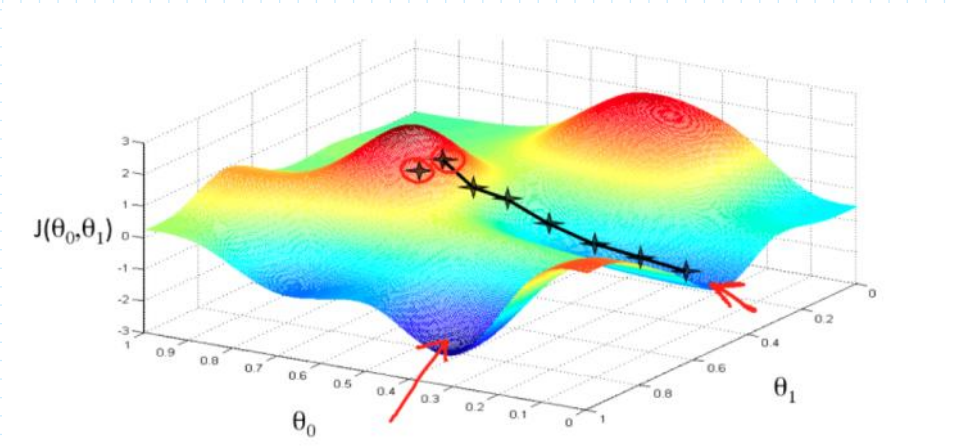


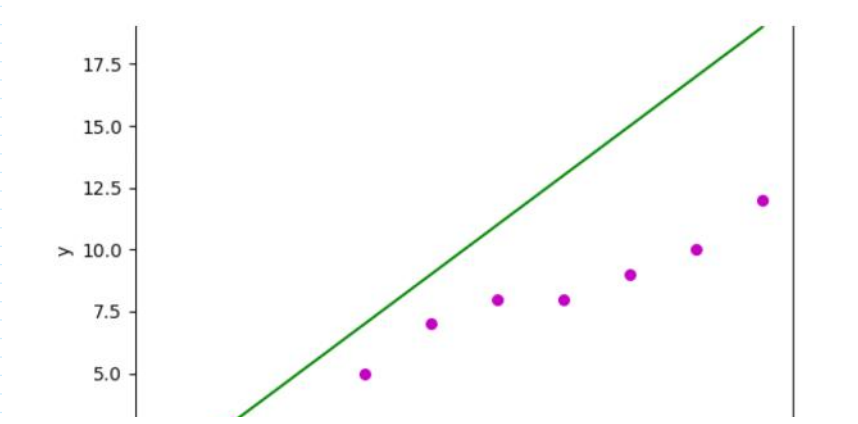
Gradient descent algorithm is an optimisation algorithm that uses to find the optimal value of parameters that minimises loss function. For instance, the algorithm iteratively adjusts the parameters such as weights and biases of the neural network to find the optimal parameters that minimise the loss function.

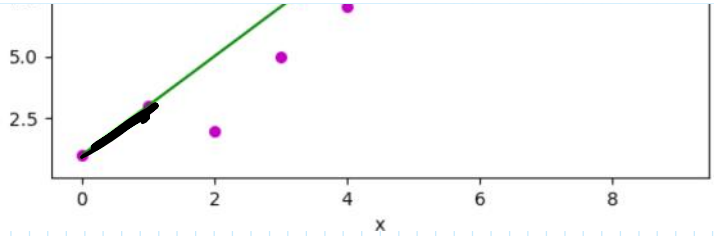


1. Initialise the coefficients m and b with random values

For example m = 1 and b =2, i.e a line equation is y = mx+b, then y = 1*x+2. This can give a line that is not best fitted for the historical data, as shown below.

$y = mx + b$





2. Compute the gradient.

We use the Sum of Squared Errors (SSE) as our loss/ cost function to minimise the prediction error. In this case, the gradient of SSE is a partial derivative of SSE w.r.t m and partial derivative of SSE w.r.t b.

SSE equation is as follows:

$$SSE = \frac{1}{2}(Y - \overline{Y})^2$$

$$\overline{Y} = mx + b$$

3. Update coefficients in the direction of optimal m and b

We can update the coefficients m and b using the gradient calculated from the above equations.

LR:

Update rule of coefficient m is:

$$m = m - r * \boxed{\frac{\partial}{\partial m} SSE}$$

Update rule of coefficient b is:

$$b = b - r * \frac{\partial}{\partial b} SSE$$

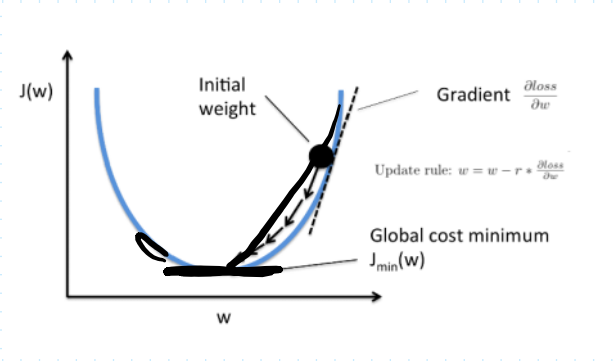
Here r is the learning rate.

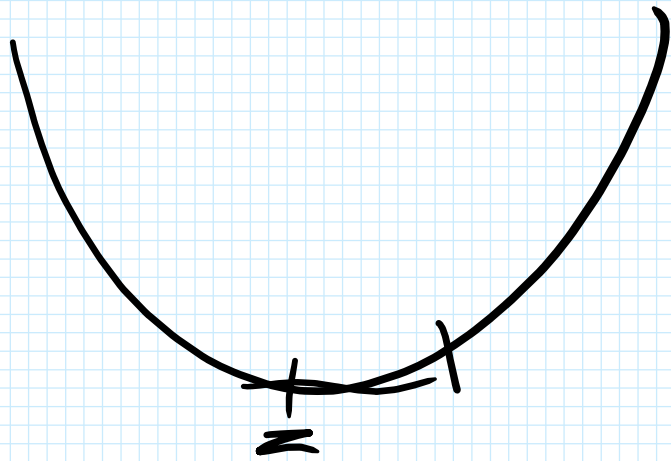
4. Use new m and b for prediction

We use the data X with new m and b, computed in the above step, to draw the line that fit the data. We calculate the SSE of each data point in X to find out the total SSE, which is a sum of squared errors of data points, divided of 2. The total SSE indicates the error rate of the model prediction.

$$Total\ SSE = \frac{1}{2}\sum_{i=1}^{n}(Y - \overline{Y})^2$$

5. Repeat steps 2, 3 and 4





Gradient descent algorithm for Linear Regression

$$h_\theta(x) = \theta_0 + \theta_1 x$$

$J(\theta_0, \theta_1)$ Cost function $= \frac{1}{2m} \sum_{i=1}^{m} \left(\hat{y}_i - y_i \right)^2$

$\hat{y}_i \rightarrow h_\theta(x_i)$

* predicted value corresponding to x_i

$J(\ ) =$ Mean Squared error function.

For best case. $\hat{y}_i = y_i$: Lines passes through all actual points

Thus $J(\theta_0, \theta_1) = 0$

GD is used to automatically improve $J(\theta_0, \theta_1)$ function

learning rate

$$\theta_j : \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) \quad \text{for } j = 0, 1$$

Repeat till it converges.

$$\theta_0 = \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_1 = \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

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after derivation & taking mean of Error.

$$\theta_j = \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} (\hat{y}_i - y_i)$$

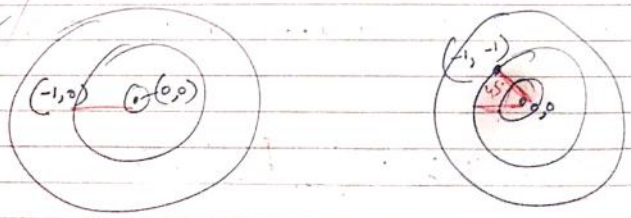
iteration

$$\theta_0^{i+1} = \theta_0^{i} - \alpha \frac{1}{m} \sum (\hat{y}_i - y_i)$$

$$\theta_1^{i+1} = \theta_1^{i} - \alpha \frac{1}{m} \sum x_i (\hat{y}_i - y_i)$$

Learn More with example.

Rotational invariance



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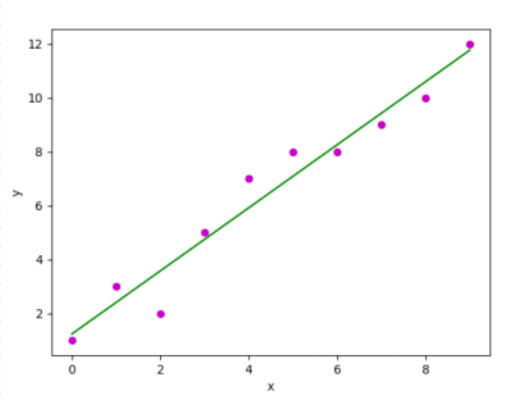

3. Multiple Regression

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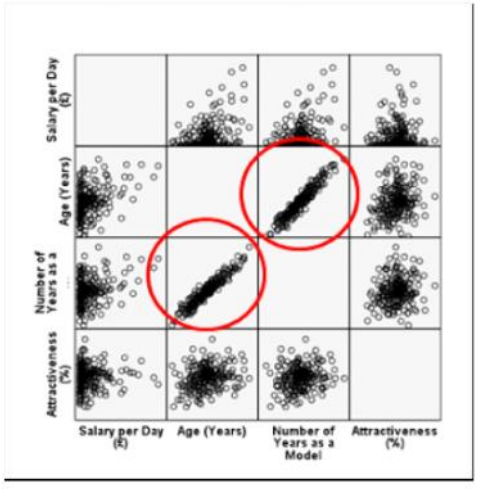
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

salary	age	years	beauty
.37	16.67	3.15	78.25
53.72	20.35	5.51	68.57
1.46	18.20	5.33	75.04
.02	15.36	3.84	65.14
95.34	24.17	8.53	71.77
14.64	18.26	4.39	78.05
8.67	17.70	4.40	72.07
2.65	17.49	4.11	75.33
7.55	17.07	3.52	72.03
1.20	20.08	6.83	71.93
10.16	22.77	7.32	86.20
19.88	21.69	6.16	76.32
.84	16.72	3.80	65.44
.16	18.39	4.66	75.13
1.49	17.32	3.83	84.70
3.60	18.38	4.83	76.94
.64	18.33	4.40	81.38
4.65	12.17	.43	83.08
2.97	18.89	5.05	80.37
25.86	17.80	4.35	87.93
.16	16.24	3.49	76.85
.76	17.20	3.95	82.30
6.42	18.99	5.24	99.22



4. Polynomial Regression

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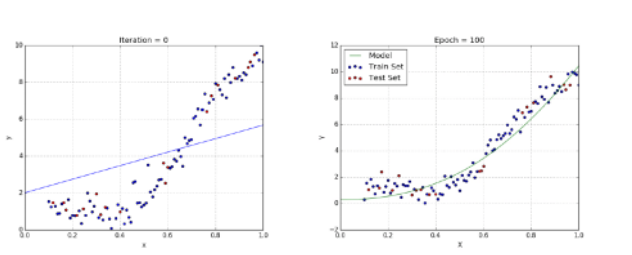

Linear vs Polynomial Regression with data that is non-linearly separable

Curvi Linear Regression. (6)

$$Y = a + bx + cx^2$$

Error $= y_i - \hat{y}_i$

Sum of square form $E = \sum_{i=1}^{n} (y_i - a - bx - cx^2)^2$

$$\frac{\partial E}{\partial a} = 0 \Rightarrow \sum y_i = na + b\sum x_i + c\sum x_i^2$$

$$\frac{\partial E}{\partial b} = 0 \Rightarrow \sum x_i y_i = a\sum x_i + b\sum x_i^2 + c\sum x_i^3$$

$$\frac{\partial E}{\partial c} = 0 \Rightarrow \sum x_i^2 y_i = a\sum x_i^2 + b\sum x_i^3 + c\sum x_i^4$$

X	Y
1	6
2	7
3	8
4	9

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Fitting of a Power Curve. (7)

$$Y = ax^b$$

$$E = (y_i - ax_i^b)^2$$

$$\log Y = \log a + b \log x$$

$$U = A + bV$$ ↘ again use lsm m.

X	Y	$U = \log x$	$V = \log y$

$$\sum U = nA + b\sum V \quad - (1)$$

$$\sum UV = A\sum V + b\sum V^2 \quad - (2)$$

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Fitting of exponential Curve. (8)

$$Y = ab^x$$

$$\log Y = \log a + x \log b$$

$$\boxed{U = A + BX}$$

$$\Sigma U = nA + B\Sigma X \quad \text{— (1)}$$

$$\Sigma XU = A\Sigma X + B\Sigma X^2 \quad \text{— (2)}$$

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9. $Y = a + bx + cx^2$ (9)

X	Y	X^2	X^3	X^4	XY	X^2Y	x
1	2						
1	7						
2	7						
2	10						
3	8						
3	12						
4	10						
5	14						
6	11						
7	14						
(A)	(B)	(C)	(D)	(E)	(F)	(G)	

$$B = na + bA + cC$$

$$F = aA + bC + cD$$

$$G = aC + bD + cE$$

$$Y = 1.8 + 3.48X - .097X^2$$

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Q Fit an exponential curve. (10)

X	Y	U = log Y	XU	X^2
1	1.0			
2	1.2			
3	1.8			
4	2.5			
5	3.6			
6	4.7			
7	6.6			
8	9.1			
(A)	(Y)	(Z)	(U)	(V)

$Y = ab^X$

$\log Y = \log a + X \log b$

$U = A + BX$

$Z = nA + B\cancel{X}$

$U = AX + BV$

$a = \text{antilog}(A)$ $\quad b = \text{antilog}(B)$

$Y = (.6821)(1.38)^x$

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$$\frac{\partial E}{\partial a} = -2\sum(y_i - a - bx_i) = 0$$

$$\sum y_i - na - b\sum x_i = 0$$

$$\sum x_i y_i - a\sum x_i - b\sum x_i^2 = 0$$

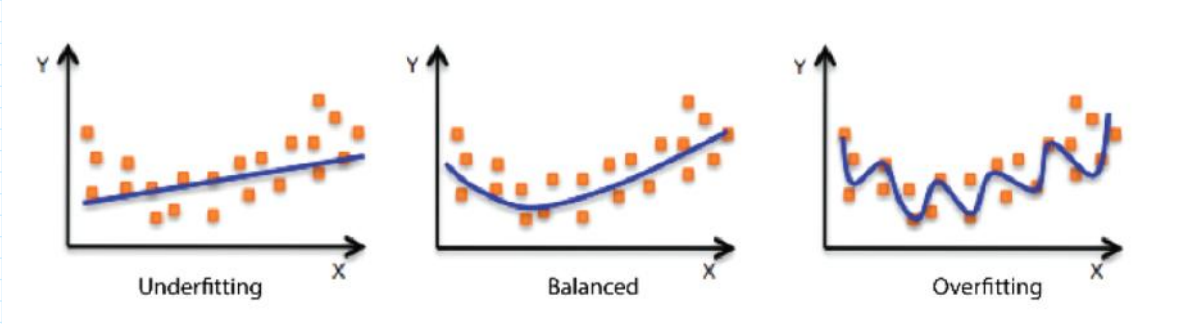
X	Y

$$\frac{\partial E}{\partial b} = -2\sum x_i(y_i - a - bx_i) = 0$$

Lasso and Ridge Regression

15 February 2021 10:39 AM

Is a model with the lowest RSS truly the best model?



BIAS v/s VARIANCE

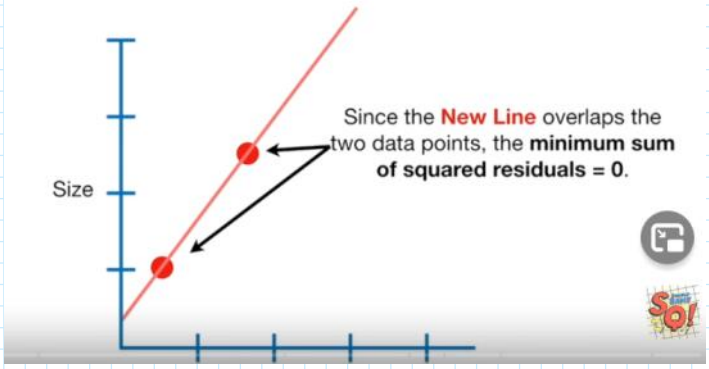
- bias is related with a model failing to fit the training set
- variance is related with a model failing to fit the testing set

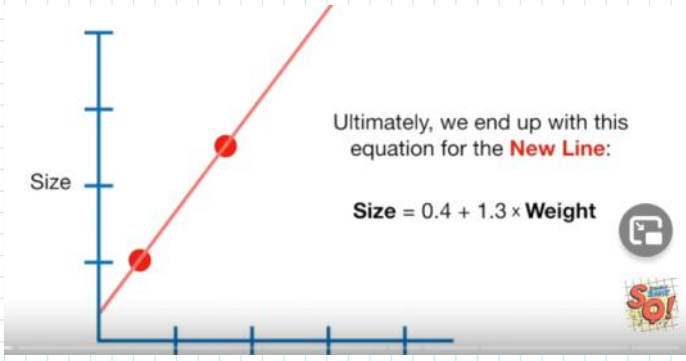
Ridge Regression

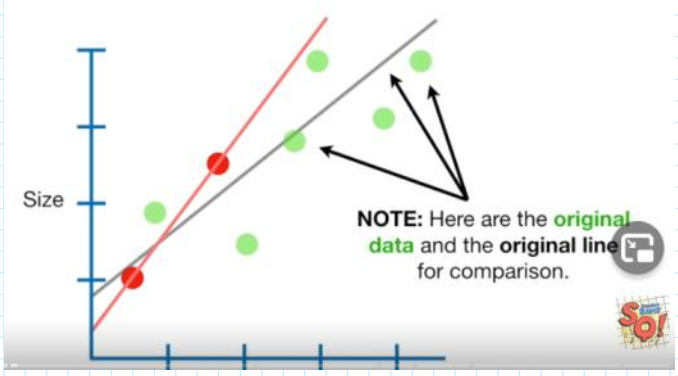
Ridge regression shrinks the regression coefficients, so that variables, with minor contribution to the outcome, have their coefficients close to zero.

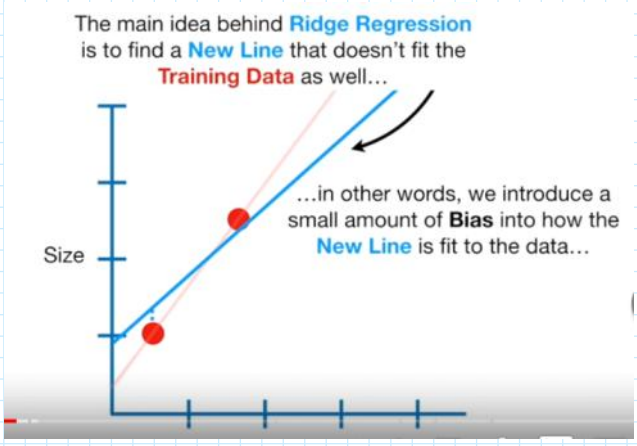
The shrinkage of the coefficients is achieved by penalizing the regression model with a penalty term called L2-norm, which is the **sum of the squared coefficients**.

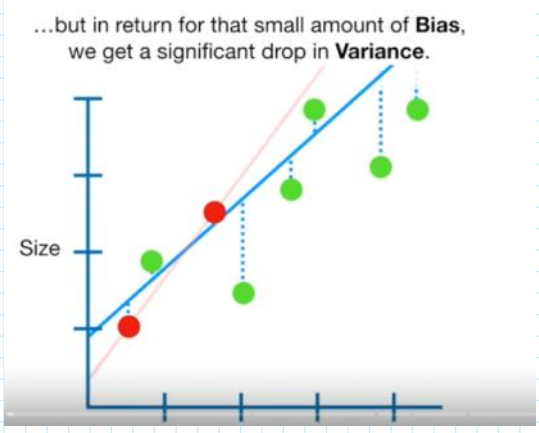
$$\sum_{i=1}^{n} (y-\hat{y})^2 + \lambda \times (\text{Slope})^2$$

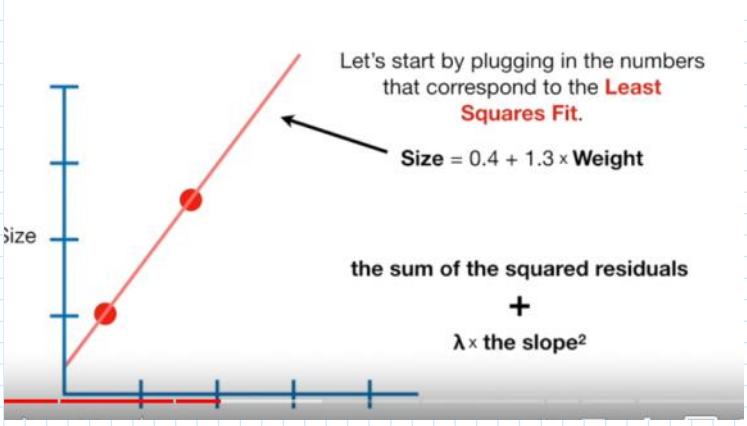

Size
Since the New Line overlaps the two data points, the minimum sum of squared residuals = 0.

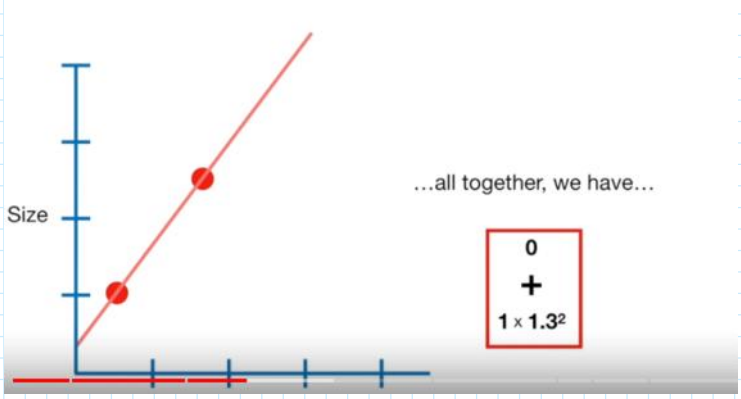

Size
Ultimately, we end up with this equation for the New Line:
Size = 0.4 + 1.3 × Weight


Size
NOTE: Here are the original data and the original line for comparison.

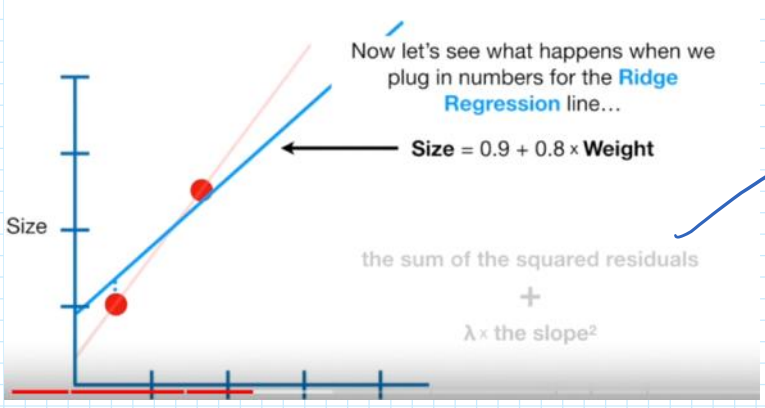

The main idea behind Ridge Regression is to find a New Line that doesn't fit the Training Data as well…
…in other words, we introduce a small amount of Bias into how the New Line is fit to the data…
Size


…but in return for that small amount of Bias, we get a significant drop in Variance.
Size





Total penalty = 1.69



$(.3)^2 + (.8)^2$

$+ \lambda \times (.8)^2$

Total penalty = .74

When λ=0, the penalty term has no effect, and ridge regression will produce the classical least square coefficients. As lambda gets larger, the bias is unchanged but the variance drops. However, as λ increases to infinite, the impact of the shrinkage penalty grows, and the ridge regression coefficients will get close to zero.

where λ*(slope)^2 is the penalty factor that shrinks the coefficients closer to zero , but not exactly zero. The amount of the penalty can be fine-tuned using a constant called lambda (λ). Selecting a good value for λ is critical.

REGULARIZATION L1 (Lasso) L2 (Ridge)

In layman term, **<u>Regularization helps the over-fitting problem of Linear Regression by shrinking the corresponding coefficients towards zero</u>** .

$$Y = (mx + c) * \left.\begin{matrix} \lambda\,|slope| \\ / \\ \lambda\,(slope)^2 \end{matrix}\right\} \text{penalty factor}$$

where λ*|slope| is Lasso(L1 penalty) , λ*(slope)^2 is Ridge(L2 penalty) Regression.

<u>LASSO(Least Absolute Shrinkage and Selection Operator)</u>

Here observe the keywords- ABSOLUTE , SELECTION (because this is what lasso does!)
Lasso shrinks the regression coefficients toward zero by penalizing the regression model with a penalty term called L1-norm, which is the <u>sum of the absolute coefficients</u>.
Here is the equation,

$$\underbrace{\sum_{i=1}^{n} (y-\hat{y})^2}_{\text{cost function}} + \underbrace{\lambda \sum_{j=1}^{m} |slope|}_{\text{shrinks exactly to zero.}}$$

The amount of the penalty can be fine-tuned using a constant called lambda (λ). Selecting a good value for λ is critical.

- Lasso is generally used when we have more number of features because it automatically does feature selection , but how?
  Suppose we have three features , Y =m1x1+m2x2+m3x3+c
  Y=λ*|m1+m2| ; suppose m3 's slope is very close to zero it will remove slope of m3.

Lasso shrinks the less important feature's slope to zero ie whenever the slope value is very less(close to

zero) those features will be removed which means that they don't have a part for predicting the best fit line.

When lambda is small, the result is essentially the same as the slope of linear regression . As lambda increases, shrinkage occurs so that variables that are at zero can be thrown away. But selecting a large lambda value causes under-fitting(Under-fitting means that you are not capturing the patterns enough) .

5. Questions

15 February 2021 10:27 AM

x_i	y_i	x_i^2	$x_i y_i$
1	2	1	2
2	5	4	10
3	4	9	12
4	8	16	32
5	6.	25	30
15	25	55	86

$$5a + 15b = 25$$

$$15a + 55b = 86$$

$$y = 1.7 + 1.1x$$

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also estimate the value of X at $Y=70$. (E)

X	Y	x_i^2	$x_i y_i$
65	67		
66	68		
67	65		
67	68		
68	72		
69	72		
70	69		
72	71		
Σx A	ΣY (B)	(C)	(D)

$A = na + bB$ $\qquad$ $D = aA + bC.$

Calculate a, b.

$$\boxed{Y = .665X + 23.78.}$$

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Coefficient of Corelation between X and Y

15 October 2020 02:51 PM

$$\rho_{XY} = \frac{Cov(XY)}{\sigma(X)\,\sigma(Y)} = \frac{\sigma(XY)}{\sigma(X)\,\sigma(Y)}$$

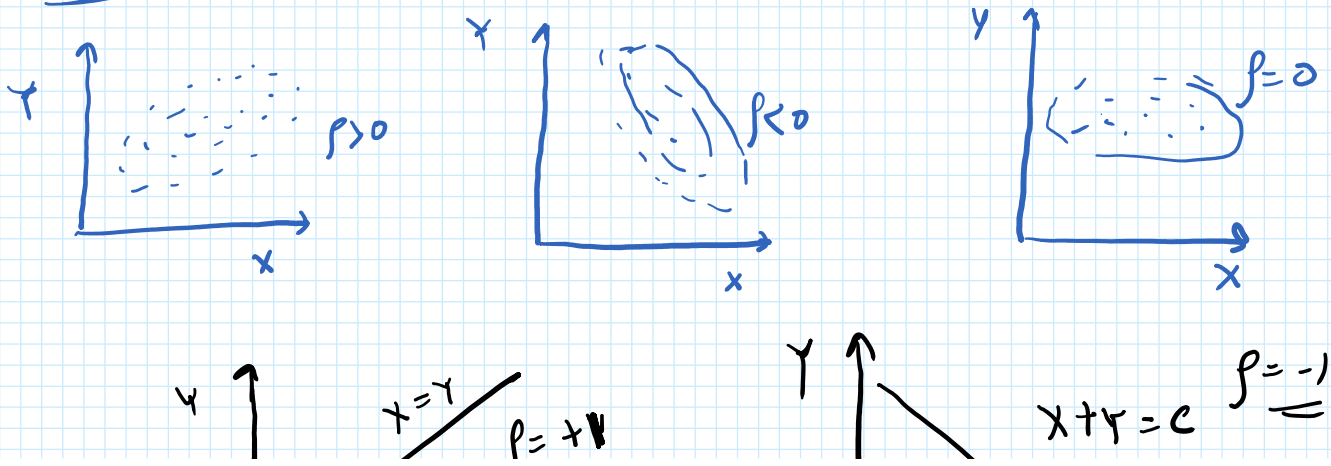
Remarks.

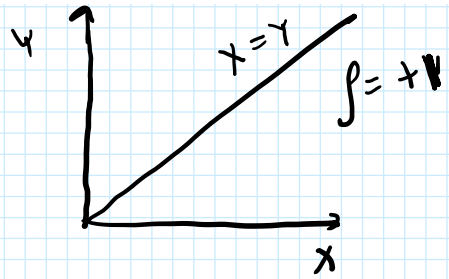
1. $-1 \le \rho_{XY} \le 1$ for any X & Y
2. It only measure <u>Linear relation</u> b/w r.v.'s.
3. If $\rho_{XY} = 0$, we say that X & Y are <u>Uncorrelated</u>.
4. It is a statistical measure that how strongly two Variables are related.
5. ρ_{XY} doesn't get affected by change of Scale in (X, Y).
6. It is a <u>unit free measure</u>.
7. if $\rho_{XY} = +ve$ then relation is direct or positive.

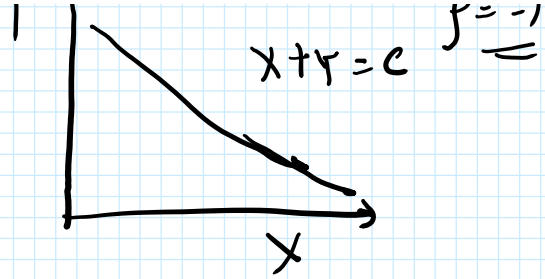
   $-ve$ " " " diverse or -ve.

$$\rho_{XY} = \left.\begin{matrix} +1 \\ -1 \end{matrix}\right] \text{Perfect} \begin{cases} +ve \\ -ve \end{cases}$$

Significance.







Limit of Correlation Coefficient

Method 1 Let.

$$U = \frac{X-\mu_X}{\sigma_X} \qquad V = \frac{Y-\mu_Y}{\sigma_Y}$$

$$E(U)=0 \qquad E(V)=0$$

$$E(U^2)=1 \qquad E(V^2)=1$$

$$E(U \pm V)^2 \geq 0$$

$$E(U-V)^2 \geq 0$$

$$E[U^2+V^2-2UV] \geq 0$$

$$E(U^2)+E(V^2)-2E(UV) \geq 0$$

$$1+1-2E(UV) \geq 0$$

$$\boxed{E(UV) \leq 1} \quad -(1)$$

$$E(U+V)^2 \geq 0$$

$$E(U^2+V^2+2UV) \geq 0$$

$$E(U^2)+E(V^2)+2E(UV) \geq 0$$

$$\boxed{E(UV) \geq 1} \quad -(2)$$

from (1) & (2)

$$-1 \leq E(UV) \leq 1$$

$$-1 \leq E\left[\frac{(X-\mu_X)}{\sigma_X}\frac{(Y-\mu_Y)}{\sigma_Y}\right] \leq 1$$

$$-1 \leq E\left[\frac{(x-\mu_x)}{\sigma_x}\frac{(Y-\mu_Y)}{\sigma_Y}\right] \leq 1$$

$$-1 \leq \rho_{XY} \leq 1$$

Theorm: if X & Y are independent then $\rho_{XY} = 0$. But converse is not true.

??

Proof using eg.

X \ Y	-1	0	1
0	0	1/3	0
1	1/3	0	1/3

$E(X) = \frac{2}{3}$ $\qquad$ $E(Y) = 0$

$E(XY) = 0$

$$\boxed{Cov(X,Y) = 0 \quad \text{then} \quad \rho_{XY} = 0}$$

Reverse:

$$p_{XY}(0,-1) = p_X(0) \cdot p_Y(-1)$$

P_{XY} (,)

$$0 = \frac{1}{3} \cdot \frac{1}{3}$$

X & Y are not independent.